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1180

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
PUNJABI UNIVERSITY, PATIALA
SUBJECT – DISCRETE MATHEMATICS (B.TECH 3SEM)

TIME -1 HOUR

MM- 15

SECTION – A

1. Find the coefficients of x^{15} in the $(2x + 3y)^{60}$. (1)
2. Let R be a relation on the set $A = \{1, 2, 3, 4\}$ defined by $R = \{(1, 2), (2, 3), (1, 3), (3, 4)\}$. Find symmetric closure. (1)
3. How many 6-digit numbers without repetition of digits are such that the digits are all non-zero and 1 and 2 do not appear consecutive in either order. (1)
4. Draw Hasse Diagram for D_{20} . (2)

SECTION – B (Do any two)

5. Solve $5s(k) - 6s(k-1) + 9s(k-2) = 3^k$. (5)
6. Let $f: R \rightarrow R$ is defined by $f(x) = ax + b$, $a, b, x \in R$ and $a \neq 0$. Show that f is invertible and find the inverse of f . (5)
7. Determine whether the relation is reflexive, symmetric, anti-symmetric or transitive given R is a relation on Z , where xRy is $(x+y)$ is odd. (5)

MM.15

Note:

Subject: Discrete Mathematics

Date: 18-09-2025

MST-1

Section-A is compulsory & attempt any two questions from

Section-B

B.Tech-Part-II (3rd sem)

Branch: 2CE,

Section-A

(5*1=5 marks)

Q1.

- Give an example of a POSET that has a least element but no greatest element.
- Find $g \circ f$ Where $f(x) = 4x + 1$ and $g(x) = x^2 - 2$
- Can a relation be both symmetric and antisymmetric? Give an example.
- Does an antisymmetric closure exist? Justify.
- From the word DISCRETE, how many distinct arrangements of letters are possible?

Section-B (Attempt any two)

(5*2=10 marks)

Q2. The Fibonacci sequence F_n is defined by the recurrence relation $F_n = F_{n-1} + F_{n-2}, n \geq 2$

With initial conditions $F_0 = 0, F_1 = 1$. Solve the recurrence relation.

Q3. Draw the HASS diagram for the POSET $(\{2, 3, 5, 6, 15, 25, 24, 150, 450, 600\}, |)$. Also find out the LUB of the subset $\{2, 3, 5\}$ and the GLB of the subset $\{24, 450\}$.

Q4. Consider the relation R on the set $N \times N$ defined as: for $(a, b), (c, d) \in N \times N$,
 $(a, b) R (c, d) \Leftrightarrow ad = bc$. Prove that it is an equivalence relation.

Section-A (1 marks each)

1. Let f be the function that assigns the last two bits of a bit string of length 2 or greater to that string. For example, $f = (11010) = 10$, find the domain, co-domain and range.
2. Let R be the set of all binary relations on the set $A = \{1, 2, 3\}$. Suppose a relation is chosen from R at random. Calculate probability that the chosen relation is reflexive. (Round off to 3 decimal places).
3. Calculate the cardinality of power set of $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. — no. of elements (2^{11})
4. Let S be a set of n elements. Find the number of ordered pairs in the largest and the smallest equivalence relations on S .
5. What is total ordering?

Section-B (5 marks each)

6. Draw the Hasse diagram for the $(\{1, 2, 3, 4, 5, 9, 900\}, |)$, how many complements does the element '2' have and find them in the same Hasse diagram.
7. Find the equivalence classes for congruence modulo 4? On set of positive integers i.e. $\mathbb{Z}^+ = \{0, 1, 2, \dots, \infty\}$
8. Consider the following Hass Diagrams (figure 1 to 4). Which of following represents a lattice and non-lattice? Give proper justification.

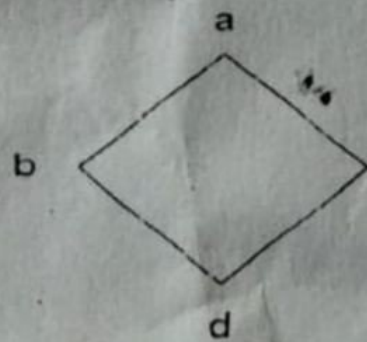


Fig:1



Fig:2

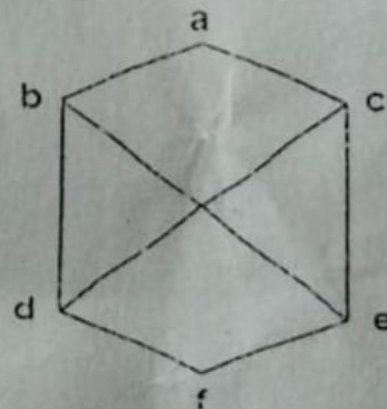


Fig:3

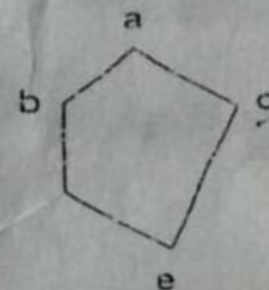


Fig:4

MM.15

MST-I

Date: 11-10-2021

Subject: Discrete Mathematics

Note: Section-A is compulsory & attempt any two questions from Section-B Time: 10.00 – 11.00

MST-I

B.Tech-Part-II (3rd sem)

Branch: 2CE,

SECTION - A (1*5=5 marks)

Q.1: a) Define POSET with example.

b) Give an example of a relation which is neither reflexive, nor symmetric, nor transitive, nor anti-symmetric?

c) Prove that $f(x)=3x+5$ for all $x \in \mathbb{R}$ is invertible function, where $f: \mathbb{R} \rightarrow \mathbb{R}$.

d) Define Lattice with an example.

e) Find $g \circ f$ Where $f(x)=4x+1$ and $g(x)=x^2-2$.

SECTION - B (5 *2=10 marks)

Q.2: Solve recurrence relation $a_n = 4a_{n-1} - 4a_{n-2} + (n+1) + 2^n$ Q.3: a) Let R be relation on the set of ordered pair of positive integers such that $(a,b), (c,d) \in R$ if and only if $a+d=b+c$. Show that R is an equivalence relation.b) Draw Hasse diagram for divisibility on D_{30} .

Q.4: a) By using recursive definition, find ten terms of Fibonacci series.

b) Find all upper bounds of $\{a, b, c\}$ and lower bounds of $\{f, g, h\}$, maximal element and least element in hasse diagram given below: